

Student Guide — primary / module-12- movement-of-sun

IKS Curriculum

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Student Workbook

Student Workbook — Movement of the Sun (Primary)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Sūrya* — _____
- *Uttarāyana* — _____
- *Dakṣiṇāyana* — _____
- *saṁkrānti* — _____
- *rāśi* — _____
- *sāyana* — _____
- *nirayana* — _____

Day-by-day notes

Day 1 — The Sun's Yearly Path

What I learned:

One question I have:

Day 2 — Axial Tilt and the Seasons

What I learned:

One question I have:

Day 3 — Uttarāyaṇa and Dakṣiṇāyana

What I learned:

One question I have:

Day 4 — Equinoxes — Viṣuva

What I learned:

One question I have:

Day 5 — Twelve Rāśis

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Day 6 — Saṁkrānti Moments

What I learned:

One question I have:

Day 7 — Sāyana vs Nirayana

What I learned:

One question I have:

Day 8 — Precession of the Equinoxes

What I learned:

One question I have:

Day 9 — Cross-Cultural Solar Calendars

What I learned:

One question I have:

Day 10 — Final Share and Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — The Sun’s Yearly Path

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Observe that sunrise direction changes; introduce the ecliptic.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *Sūrya* (IAST: Sūrya; lit. “the Sun”)
- The Sun rises in different places through the year!

Stand at a window facing east. Mark where the Sun comes up TODAY. (Just outside the window — point with arms.)

In summer, the Sun rises a bit more to the LEFT. In winter, more to the RIGHT (in northern places).

Children draw a window with the Sun in 3 spots: summer (left), middle (spring/autumn), winter (right). One drawing.

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 2 — Axial Tilt and the Seasons

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Demonstrate with torch+globe why we have summer and winter.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Uttarāyaṇa* (IAST: Uttarāyaṇa; lit. “northward solar course (Dec 21 – June 21)”)
- Why do we have seasons?

Demo: hold a ball (Earth) tilted to one side. Shine a torch (Sun) at it. Move the ball around the torch. - Sometimes the TOP of the ball gets more light → summer. - Sometimes the BOTTOM gets more → other hemisphere's summer.

“*Earth is TILTED. That's why we have seasons!*” Children draw a tilted Earth with the Sun.

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

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Day 3 — Uttarāyana and Dakṣiṇāyana

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Identify the dates and direction of each.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Dakṣiṇāyana* (IAST: Dakṣiṇāyana; lit. “southward solar course (June 21 – Dec 21)”)
 - **Uttarāyana** (“Going North”): Dec to June — days get longer in the north.
 - **Dakṣiṇāyana** (“Going South”): June to Dec — days get shorter in the north.

Children stand with hands held high. Drift to the right (north). “*That's Uttarāyana!*” Drift to the left (south). “*That's Dakṣiṇāyana!*”

Each child draws an arrow on a calendar showing today's direction.

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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Day 4 — Equinoxes — Viṣuva

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Identify spring and autumn equinoxes (day = night).

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *saṁkrānti* (IAST: saṁkrānti; lit. “Sun's crossing from one Rāśi to the next”)
- Two special days each year: day and night are EXACTLY the same length!

These are called **equinoxes** (or *viṣuva*). - **Spring equinox** (around March 21): days getting longer. - **Autumn equinox** (around September 22): days getting shorter.

Children stand still with arms balanced equally. “*Day equals night!*”

Draw two suns on a year-circle, one in March, one in September. Label “equinox.”

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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Day 5 — Twelve Rāśis

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Name the 12 rāśis in Sanskrit and pair with Greek zodiac names.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *rāśi* (IAST: rāśi; lit. “1/12th of the ecliptic; zodiacal sign”)
- The Sun visits 12 special “houses” each year!

Names (sing-chant): “*Meṣa, Vṛṣabha, Mithuna, Karka, Siṃha, Kanyā, Tulā, Vṛścika, Dhanus, Makara, Kumbha, Mīna!*”

Each “house” has a Sanskrit name AND a Greek name: - Meṣa = Aries (ram!), Vṛṣabha = Taurus (bull!), Mithuna = Gemini (twins!)...

Children pick one rāśi to draw. Today’s question: “*What’s YOUR birth-month rāśi?*”

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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Day 6 — Saṃkrānti Moments

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Identify Makara, Karka, and other saṃkrāntis on the calendar.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today's idea (10 min)

- *sāyana* (IAST: sāyana; lit. “tropical zodiac (linked to seasons)”)
- When the Sun moves from one rāṣi-house to the next, we call it *San̥krānti*.
- The most famous: *Makara San̥krānti* — when the Sun enters Makara (Capricorn). Around January 14 in many years!

Show a wheel with 12 houses. Children put a Sun-sticker on each rāṣi, then circle the date when the Sun enters Makara.

“In Tamil Nadu, this is Pongal. In Punjab, Lohri. In many places, it's a happy festival!”

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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Day 7 — Sāyana vs Nirayana

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Distinguish tropical and sidereal zodiacs; introduce ayanāṁśa.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *nirayana* (IAST: nirayana; lit. “sidereal zodiac (linked to fixed stars)”)
- There are TWO ways to measure the Sun’s path:
 - **Sāyana** (modern science): starts at the Vernal Equinox (Mar 21).
 - **Nirayana** (Indian tradition): starts at a fixed star.

These two ways DRIFT apart slowly — about 1° every 70 years. Over many centuries, they’re now ~24 days apart.

Children draw two suns: one labelled “tradition” and one labelled “modern.” They’re at slightly different spots.

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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- If energy is low, swap the activity for free drawing.

Day 8 — Precession of the Equinoxes

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Explain why Indian and Western zodiac dates differ today.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: "What's one thing you remember from yesterday?"

Today's idea (10 min)

- *Sūrya* (IAST: Sūrya; lit. "the Sun")
- The Earth wobbles slowly, like a spinning top.
- One wobble takes about 26,000 years!

Show a spinning top toy. Let it slow down — it wobbles. "*Earth does this too, just SUPER slowly.*"

Children draw Earth tilted, with a curved arrow showing the wobble. "*Astronomers from long ago noticed this!*"

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

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Day 9 — Cross-Cultural Solar Calendars

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Compare with Egyptian, Roman, Gregorian.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Uttarāyaṇa* (IAST: Uttarāyaṇa; lit. “northward solar course (Dec 21 – June 21)”)
- Many people use the Sun to make calendars!
- Egypt: 365 days, like ours.
- India: months named after the rāśis.
- Roman/modern: January, February, ...

Children write today's date in 3 ways: 1. Modern (Jan 25, 2026) 2. Tamil (rāśi-based) 3. Just by season (winter/late winter)

Discuss which they like.

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 10 — Final Share and Assessment

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Project share + summative quiz.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *Dakṣiṇāyana* (IAST: Dakṣiṇāyana; lit. “southward solar course (June 21 – Dec 21)”)
- Final day! We celebrate the Sun’s yearly journey.

Each child shows their year-wheel. Cheer for each.

Sing the 12 rāśi names together one more time.

Quiz with stickers — everyone gets a “Sun Watcher” sticker. “*Never look at the Sun! Use eyes-down only.*”

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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- If energy is low, swap the activity for free drawing.

Homework

Homework — Movement of the Sun (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — The Sun's Yearly Path** · Draw about today's concept. (~5 min)
2. **Day 2 — Axial Tilt and the Seasons** · Draw about today's concept. (~5 min)
3. **Day 3 — Uttarāyaṇa and Dakṣiṇāyana** · Draw about today's concept. (~5 min)
4. **Day 4 — Equinoxes — Viṣuva** · Draw about today's concept. (~5 min)
5. **Day 5 — Twelve Rāsis** · Draw about today's concept. (~5 min)
6. **Day 6 — Saṁkrānti Moments** · Draw about today's concept. (~5 min)
7. **Day 7 — Sāyana vs Nirayana** · Draw about today's concept. (~5 min)
8. **Day 8 — Precession of the Equinoxes** · Draw about today's concept. (~5 min)
9. **Day 9 — Cross-Cultural Solar Calendars** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Share and Assessment** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.